

ARTIFICIAL INTELLIGENCE – ASSESSMENT TOOL 2

Problem Statements

Question 1

Topic:

*Emergency Medicine Delivery using Greedy Best-First Search and A Search**

Problem Statement

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic, where roads may suddenly become inaccessible due to flooding and weather conditions influence travel cost. The drone has access to heuristic estimates based on straight-line distance, receives partial real-time updates regarding blocked paths, and experiences variable movement costs depending on terrain and weather. Because emergency response is critical, the drone must rapidly determine the safest and shortest route while minimizing delivery delay and operational risk.

Question 2

Topic:

Smart Traffic Signal Optimization using Hill Climbing and Simulated Annealing / Genetic Algorithm

Problem Statement

A smart city implements an Artificial Intelligence system to optimize traffic signal timings across hundreds of intersections. The primary objectives are to minimize total waiting time, reduce fuel consumption, and decrease traffic congestion. The system continuously adjusts signal timings according to changing traffic flow. Since the search space is extremely large and traffic conditions change dynamically, the optimization algorithm may become trapped in locally optimal solutions, requiring advanced optimization techniques to identify better signal configurations.

Question 3

Topic:

Autonomous Mars Rover using Online Search Agent

Problem Statement

An autonomous rover is deployed on the surface of Mars to explore unknown terrain and collect scientific samples. The rover must safely navigate through hazardous regions containing rocks and craters while operating with limited sensing capability and without a complete map of its environment. As it explores, the rover continuously updates its knowledge and makes real-time decisions based on newly observed information. The problem requires an intelligent online search strategy capable of handling uncertainty and dynamic environmental changes.

Question 4

Topic:

University Examination Timetable Generation using Constraint Satisfaction Problem (CSP)

Problem Statement

A university plans to develop an Artificial Intelligence system that automatically generates examination timetables. The system must satisfy multiple constraints, including preventing examination overlaps for students, managing limited examination halls, scheduling prerequisite subjects before dependent subjects, considering faculty availability, and providing special accommodations for eligible students. As the number of courses and students increases, the scheduling complexity grows significantly, making the problem suitable for formulation as a Constraint Satisfaction Problem solved using backtracking and constraint propagation techniques.

Question 5

Topic:

Strategic Game AI using Minimax and Alpha-Beta Pruning

Problem Statement

A gaming company is developing an Artificial Intelligence opponent for a real-time strategy game. The AI must compete against human players by making intelligent decisions within strict time constraints while evaluating complex game states that contain numerous possible moves. Since the game tree grows exponentially with every move, exhaustive computation becomes impractical. The challenge is to design an efficient game-playing algorithm using Minimax and Alpha-Beta Pruning that balances optimal decision-making with real-time performance requirements.